Chapter 10

Percentage

The term per cent means 'for every hundred'. It can be defined as follows:
"A per cent is a fraction whose denominator is 100 and the numerator of the fraction is called the rate per cent."
Per cent is denoted by the sign '%'.

10.1 Formula to Calculate Per Cent

If we have to find y% of x, then

$$y\% \text{ of } x = x \times \frac{y}{100}$$

Some Quick Results

- 5% of a number $= \frac{\text{Number}}{20}$
- 10% of a number $= \frac{\text{Number}}{10}$
- $12\frac{1}{2}\%$ of a number $= \frac{\text{Number}}{8}$
- 20% of a number $= \frac{\text{Number}}{5}$
- 25% of a number $= \frac{\text{Number}}{4}$
- 50% of a number $= \frac{\text{Number}}{2}$

Ex. 1 20% of 300 = ?
Sol. According to the formula

$$20\% \text{ of } 300 = 300 \times \frac{20}{100} = 3 \times 20 = 60$$

Alternate Method
Here, Number = 300

$$20\% \text{ of } 300 = \frac{\text{Number}}{5} = \frac{300}{5} = 60$$

Ex. 2 If 30% of $a = 60$, then find a.
Sol. According to the formula,

$$30\% \text{ of } a = 60$$
$$\Longrightarrow \quad a \times \frac{30}{100} = 60$$
$$\Longrightarrow \quad a = \frac{60 \times 100}{30} = 200$$

Conversion of Per Cent into Fraction

Expressing per cent (x%) into fraction.

$$\text{Required Fraction} = \frac{x}{100}$$

Ex. 3 Express 25% in fraction.

Sol. $$25\% = \frac{25}{100} = \frac{1}{4}$$

Ex. 4 Express 84% in fraction.

Sol. $$84\% = \frac{84}{100} = \frac{21}{25}$$

Conversion of Fraction into Percentage

Expressing a fraction $\left(\frac{x}{y}\right)$ in percent.

$$\text{Required percent } = \left(\frac{x}{y} \times 100\right)\%$$

Ex.5 Convert $\frac{3}{8}$ into percent.

Sol. Required percent $= \left(\frac{3}{8} \times 100\right)\% = 37.5\%$

Ex.6 Convert $2\frac{1}{4}$ into percent.

Sol. Required percent $= \left(2\frac{1}{4} \times 100\right)\% = \left(\frac{9}{4} \times 100\right)\% = 225\%$

Expressing One Quantity as a Per Cent with Respect to Other

To express a quantity as a per cent with respect to other quantity following formula is used.

$$\frac{\text{The quantity to be expressed in percent}}{\text{2nd quantity(in respect of which the percent has to be obtained)}} \times 100\%$$

Note: To apply this formula, both the quantities must be in same metric unit.

Ex. 7 60 kg is what per cent of 240 kg?

Sol. According to the formula,

Required Percentage

$$= \frac{\text{The quantity to be expressed in percent}}{\text{2nd quantity(in respect of which the percent has to be obtained)}} \times 100\%$$

$$= \frac{60}{240} \times 100\% = \frac{100}{4}\% = 25\%$$

Ex. 8 10 g is what per cent of 1 kg?

Sol. Here, units of both the quantities are different.

Hence, first, we will make both the units same.

1st quantity $= 10g$; 2nd quantity $= 1kg = 1000g$

Now, both quantities are in the same unit.

According to the formula,

$$\text{Required percent } = \frac{10}{1000} \times 100\% = 1\%$$

Conversion of Per Cent into Decimal

Expressing percent(x%) in decimal.

$$\text{Required decimal} = \frac{x}{100}$$

Ex. 9 Express 9% in decimal

Sol. $\text{Required decimal} = \frac{9}{100} = 0.09$

Ex. 10 Express 18% in decimal

Sol. $\text{Required decimal} = \frac{18}{100} = 0.18$

10.2 Fast Track Techniques to solve the QUESTIONS

Technique 1

If x% of A is equal to y% of B, then

$$z\% \text{ of } A = \left(\frac{yz}{x}\right)\% \text{ of } B$$

Ex. 11 If 10% of A is equal to 12% of B, then 15% of A is equal to what per cent of B?

Sol. Given that, $x = 10$, $y = 12$ and $z = 15$

$$\therefore \qquad 15\% \text{ of A} = \left(\frac{12 \times 15}{10}\% \text{ of B}\right) = 18\% \text{ of B}$$

Technique 2

When a number x is increased or decreased by y%, then the new number will be

$$\frac{100 \pm y}{100} \times x$$

Note

1. '+' sign is used in case of increase
2. '-' sign is used in case of decrease

Ex. 12 The monthly income of a person is ₹8000. If his income is increased by 20%, then what will be his new monthly income?

Sol. Monthly income of a person = ₹8000

$$\text{Increment in income} = 8000 \times \frac{20}{100} = 1600$$

$$\text{New income} = 8000 + 1600 = ₹9600$$

Fast Track Method

Here, x = ₹8000 and $y = 20\%$

According to the formula,

$$\text{New Income} = \frac{100 + 20}{100} \times 8000 \qquad \text{['+' sign is used for increase in income]}$$

$$= \frac{120}{100} \times 8000 = ₹9600$$

Ex. 13 The price of a computer is ₹20000. What will be the price of computer after reduction of 25%?

Sol. Here, $x = ₹20000$ and $y = 25\%$

According to the formula,

$$\text{New Price} = \frac{100-25}{100} \times 20000$$
$$= \frac{75}{100} \times 20000 = 75 \times 200 = ₹15000$$

Technique 3

❒ If x is $a\%$ more than y, then y is $\left(\frac{a}{100+a}\right) \times 100\%$ less than x

❒ If x is $a\%$ less than y, then y is $\left(\frac{a}{100-a}\right) \times 100\%$ more than x

Ex. 14 If income of Ravi is 20% more than that of Ram, then income of Ram is how much percent less than that of Ravi?

Sol. Let income of Ram be 100

Then Ravi's income $= 120$

$\therefore$ The Required Percentage $= \frac{120-100}{100} \times 100\% = \frac{20}{120} \times 100\% = 16\frac{2}{3}\%$

Fast Track Method

Here, $a = 20\%$

According to the formula,

$\therefore$ Required Percentage $= \left(\frac{a}{100+a}\right) \times 100\%$

$$= \left(\frac{20}{100+20}\right) \times 100\% = \frac{50}{3}\%$$

Alternate Method

Ravi's income=20% more than that of Rams=$\frac{1}{5}$ more than that of Ram

i.e, 1 out of 5 part is more than Ravi have, Hence we can say Ravi's income is 6. If Ram's income is 5 and 1 part is less in the salary of Ram as compared to Ravi.

$$\text{Lesser\%} = \frac{\text{Part which is less in salary of Ram}}{\text{Ravi's Salary}} \times 100$$

$$= \frac{1}{6} \times 100 = 16\frac{2}{3}\%$$

Ex. 15 If in an examination, the marks obtained by Preeti is 20% less than that of Vandana, then marks obtained by Vandana is how much per cent more than marks obtained by Preeti?

Sol. Here, $a = 20\%$

According to the formula,

$$\therefore \quad \text{Required Percentage} = \left(\frac{a}{100-a}\right) \times 100\%$$

$$= \left(\frac{20}{100-20}\right) \times 100\% = \left(\frac{20}{80}\right) \times 100\% = 25\%$$

Technique 4

If the value of a number is first increased by $a\%$ and later decreased by $a\%$, then the net effect is always a decrease which is equal to $a\%$ of a and is written as

$$\frac{a^2}{100}\% \qquad \text{or} \qquad \left(\frac{a}{10}\right)^2\%$$

Ex. 16 The salary of a worker is first increased by 5% and then it is decreased by 5%. What is the change in his salary?

Sol. Let the initial salary of workers be ₹100.

Firstly, the salary of the workers is increased by 5%.

$$\text{So increased salary } = 105\% of 100 = \frac{105 \times 100}{100} = |105$$

Now, the salary is reduced by 5% after the increase

$$\therefore \qquad \text{Reduced salary} = 95\% of 105 = \frac{95 \times 105}{100} = ₹99.75$$

$\therefore$ Required change is a decrease.

$$i.e, 100 - 99.75 = 0.25$$

$$\text{So, required percentage decrease in salary} = \frac{0.25 \times 100}{100}\% = 0.25\%$$

Fast Track Method

Here, $a = 5\%$

According to the formula,

$$\text{Decreased Percentage} = \frac{a^2}{100}\%$$

$$= \frac{5^2}{100}\% = \frac{25}{100}\% = 0.25\%$$

Technique 5

When the value of an object is first changed (increased or decreased) by $a\%$ and then changed (increased or decreased) by $b\%$, then

$$\text{Net affect} = \left[\pm a \pm b + \frac{(\pm a)(\pm b)}{100} \right]\%$$

Net effect is a increase or a decrease according to the +ve or -ve sign, respectively of the final result.

Note

1. Signs of a and b depends on increment and decrement of the quantity. '+' sign is used for increment and '-' sign for decrement.
2. The above formula can also be used to find net change in the product of two numbers, if they are increased or decreased by $a\%$ and $b\%$.

Ex. 17 The price of an article is first increased by 20% and later on the price were decreased by 25% due to reduction in sales. Find the net percentage change in final price of Article.

Sol. Let the original price od article be ₹100.

Then after the first change *i.e.*, increase of 20%

$$\text{The Price of article } = \frac{100+20}{100} \times 100 = ₹120$$

Again, the price were decreased by 25%, so the reduced price

$$= \frac{100-25}{100} \times 120 = \frac{75 \times 120}{100} = ₹90$$

$\therefore$ Required decrease $= 100 - 90 = ₹10$ [$\because$ final price < initial price]

Fast Track Method

Here, $a = 20\%$, $b = 25\%$

$$\text{So, required percentage} = \frac{10}{100} \times 100 = 10\%$$

$$\text{Required change} = \left[\pm a \pm b + \frac{(\pm a)(\pm b)}{100} \right]\%$$

$$= \left[20 - 25 + \frac{20 \times (-25)}{100} \right]\% = [-5-5]\% = -10\%$$

$\therefore$ Net percentage change is a decrease of 10% because final result is negative.

Technique 6

If the price of a commodity increases or decreases by a%, then the decrease or increase in consumption, so as not to increase or decrease the expenditure is equal to

$$\left(\frac{a}{100 \pm a}\right) \times 100\%$$

Ex. 18 If the price of a commodity be raised by 40%, by how much per cent must a householder reduce his consumption of that commodity, so as not to increase his expenditure?

Sol.Let initial price be ₹100

$$\text{Price after increase} = 140\% \text{ of } 100 = \frac{140 \times 100}{100} = ₹140$$

$$\therefore \quad \text{Required reduction} = \frac{140-100}{140} \times 100\%$$

$$= \frac{40}{140} \times 100\% = 28\frac{4}{7}\%$$

Fast Track Method

Here, $a = 40\%$

According to the formula,

$$\text{Reduction in consumption} = \left(\frac{40}{100+40} \times 100\right)\%$$

$$= \left(\frac{40}{140} \times 100\right)\% = \frac{200}{7}\% = 28\frac{4}{7}\%$$

Ex. 19 If the price of milk falls down by 20%, by how much per cent must a householder increase its consumption, so as not to decrease his expenditure on this item?

Sol. Here,$a = 20\%$

According to the formula,

$$\text{Increase in consumption} = \left(\frac{20}{100-20} \times 100\right)\%$$

$$= \left(\frac{20}{80} \times 100\right)\% = 25\%$$

Technique 7

The passing marks in an examination is P%. If a candidate scores R marks and fails by F marks, then

$$\text{Maximum marks, } M = \frac{100(R+F)}{P}$$

Ex. 20 A student has to score 30% marks to get through. If he gets 30 marks and fails by 30 marks, then find the maximum marks set for the examination.

Sol. Let the maximum marks be x.

$$30\% \text{ of } x = 30 + 30$$

$$\implies \quad \frac{30x}{100} = 60 \implies x = 200$$

Fast Track Method

Here, $P = 30,\ R = 30$ and $F = 30$

According to the formula,

$$\text{Maximum Marks} = \frac{100(R+F)}{P}$$

$$= \frac{100(30+30)}{30} = \frac{100 \times 60}{30} = 200$$

Technique 8

A candidate scores x% marks in an examination and fails by a marks, while an another candidate who scores y% marks, gets b marks more than the minimum required passing marks. The maximum marks for the examination is given as

$$M = \frac{100(a+b)}{y-x} = \frac{\text{Sum of scores}}{\text{Difference in \% marks}} \times 100$$

Ex. 21 A candidate scores 25% and fails by 60 marks, while an another candidate who scores 50% marks, gets 40 marks more than the minimum required marks to pass the examination. Find the maximum marks for the examination.

Sol. Let maximum marks be x.

Marks scored by first candidate + 60 = Marks scored by second candidate - 40

$$\implies \quad 25\% \text{ of } x + 60 = 50\% \text{ of } x - 40$$

$$\implies \quad \frac{25x}{100} + 60 = \frac{50x}{100} - 40$$

$$\implies \quad \frac{25x}{100} = 100$$

$$\implies \quad x = 400$$

Fast Track Method

Here, $x = 25,\ y = 50,\ a = 60$ and $b = 40$

According to the formula,

$$M = \frac{100(a+b)}{y-x} = \frac{100(60+40)}{50-25} = \frac{100 \times 100}{25} = 400$$

Technique 9

Suppose in an examination, x% of total number of students failed in subject A and y% of total number of students failed in subject B and z% failed in both the subjects. Then,

(i) Percentage of students who passed in both the subjects= $[100-(x+y-z)\%$

(ii) Percentage of students who failed in either subject = $(x+y-z)\%$

Ex.22 In an examination, 20% of total number of students failed in History, 15% of total number of students failed in Hindi and 5% of total number of students failed in both. Find the percentage of students who passed in both the subjects.

Sol. Let total number of students be 100.

Then, students failed in History only $= 20-5=15$

And students failed in Hindi only $= 15-5=10$

$\therefore$ Total number of failed students $= 15+10+5=30$

$\therefore$ Number of students passed in both subjects $= 100-30=70$

$\therefore$ Required percentage = 70%

Fast Track Method

Here, $x=20$, $y=15$ and $z=5$

According to the formula,

Required percentage $= [100-(x+y-z)]\%$

$= [100-(20+15-5)]\% = (100-30)\% = 70\%$

Technique 10

If due to r% decrease in the price of an item, a person can buy A kg more with ₹x. then

$$\text{Actual price of that item} = ₹\frac{rx}{(100-r)A}\text{per kg}$$

Ex. 23 If due to 10% decrease in the price of sugar, Ram can buy 5 kg more sugar with ₹100, then find the find actual price of sugar.

Sol. Let the actual price of sugar be ₹y.

$$\text{Amount of sugar bought in ₹100} = \frac{100}{y}$$

$$\text{Price of sugar after 10\% decrease} = 90\% \text{ of } y = \frac{9}{10}y$$

$$\text{Now, amount of sugar bought in ₹100} = \frac{1000}{9y}$$

According to the question,

$$\frac{1000}{9y} - \frac{100}{y} = 5$$

$$\Rightarrow \quad \frac{1000-900}{9y} = 5$$

$$\Rightarrow \quad y = \frac{100}{9\times 5}$$

$$\therefore \quad y = \frac{20}{9} = ₹2\frac{2}{9}$$

Fast Track Method

Here, $r = 10\%$, $x = ₹100$ and $A = 5\ kg$

$$\therefore \quad \text{Actual price of sugar} = \frac{r\times x}{(100-5)A}$$

$$= \frac{10\times 100}{(100-10)\times 5} = \frac{1000}{450} = ₹2\frac{2}{9}$$

Technique 11

If two candidates contested in an election and one candidate got x% of total votes casted and still lose by y votes, then

$$\text{Total Number of votes casted} = \frac{100\times y}{100-2x}$$

Ex. 24 In an election contested by two candidates, one candidate got 30% of total votes and still lost by 500 votes, then find the total number of votes casted.

Sol. Let total number of votes casted be x.

$$\text{Number of votes got by first candidate} = \frac{30}{100}x$$

$$\text{Number of votes got by second candidate} = \frac{70}{100}x$$

According to the question,

$$\text{Difference in votes} = 500$$

$$\Rightarrow \quad \frac{70x}{100} - \frac{30x}{100} = 500$$

$$\Rightarrow \quad \frac{40x}{100} = 500$$

$$\therefore \quad x = \frac{500\times 100}{40} = 1250$$

Fast Track Method

Here, $x = 30$, and $y = 500$

$\therefore$ Total number of votes $= \dfrac{100 \times y}{(100 - 2x)}$

$= \dfrac{500 \times 100}{(100 - 2 \times 30)} = \dfrac{500 \times 100}{40} = 1250$

Technique 12

If the population of a town is P and it increases (or decreases) at the rate of R% per annum, then

(i) Population after n year $= P\left(1 \pm \dfrac{R}{100}\right)^n$

(ii) Population n year ago $= \dfrac{P}{\left(1 \pm \dfrac{R}{100}\right)^n}$

Note: Use '+' sign for increment and '-' sign for decrement

Ex.25 The population of a town is 352800. If it increases at the rate of 5% per annum, then what will be its population 2 yr hence. Also, find the population 2 yr ago.

Sol. Given that, $P = 352800$, $R = 5\%$ and $n = 2$

According to the formula,

Population after 2yr $= P\left(1 \pm \dfrac{R}{100}\right)^n$

$= 352800 \times \left(1 + \dfrac{5}{100}\right)^2 = 352800 \times \left(\dfrac{100+5}{100}\right)^2$

$= 352800 \times \dfrac{21}{20} \times \dfrac{21}{20} = 388962$

Population 2 yr ago $= \dfrac{P}{\left(1 \pm \dfrac{R}{100}\right)^n}$

$= \dfrac{352800}{\left(1 + \dfrac{5}{100}\right)^2} = 352800 \times \dfrac{20}{21} \times \dfrac{20}{21} = 320000$

Students may also solve problems through simple increasing-decreasing method of percentage because 5% increase for 2 yr means population after 2 yr, which is equal to $352800 \times \dfrac{105}{100} \times \dfrac{105}{100}$. In case, whe population decrease at the rate of 5%, then population after 2 yr is equal to $352800 \times \dfrac{95}{100} \times \dfrac{95}{100}$.

Technique 13

If the present population of a city is P and there is a increment or decrement of $R_1\%, R_2\%$ and $R_3\%$ in first, second and third year respectively, then

$$\text{Population of city after 3 yr } = P\left(1 \pm \frac{R_1}{100}\right)\left(1 \pm \frac{R_2}{100}\right)\left(1 \pm \frac{R_3}{100}\right)$$

Ex. 26 Population of a city in 2004 was 1000000. If in 2005 there is an increment of 15%, in 2006 there is a decrement of 35% and in 2007 there is an increment of 45%, then find the population of city at the end of year 2007.

Sol. Given that, $P = 1000000$ $R_1 = 15\%, R_2 = 35\%$ (decrease) and $R_3 = 45\%$

Population of city at the end of year 2007

$$= P\left(1 + \frac{R_1}{100}\right)\left(1 - \frac{R_2}{100}\right)\left(1 + \frac{R_3}{100}\right)$$

$$= 1000000\left(1 + \frac{15}{100}\right)\left(1 - \frac{35}{100}\right)\left(1 + \frac{45}{100}\right)$$

$$= 1000000 \times \frac{115}{100} \times \frac{65}{100} \times \frac{145}{100} = 1083875$$